

An efficient removal of power-line interference and baseline wander from ECG signals by employing Fourier decomposition technique

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ABSTRACT

Baseline wander (BW) and power-line interference (PLI) tend to occur in every recorded electrocardiogram (ECG) signal and can significantly deteriorate the quality of the signal. They need to be separated from the ECG signal to facilitate an accurate diagnosis of the patient. In this paper, we propose a new methodology based on the Fourier decomposition method (FDM) to separate both BW and PLI simultaneously from the recorded ECG signal and obtain clean ECG data. The proposed method employs either of discrete Fourier transform (DFT) or discrete cosine transform (DCT) in order to process the signal. Key DFT/DCT coefficients relating to BW and PLI are identified and then suppressed using optimally designed FDM based on a zero-phase filtering approach. The effectiveness of our method is validated on the MIT-BIH Arrhythmia database. Simulation results clearly demonstrate that the proposed method performs superior in comparison to the existing state-of-the-art techniques at different levels of signal to noise ratio power (SNR). Moreover, this method has low computational complexity which makes it suitable for real-time pre-processing of ECG signals.

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1. Introduction

Electrocardiogram (ECG) signal is a representation of heart muscles' electro-physiologic process of polarizing and depolarizing. It carries information about the heart's structure and the functioning of its electric conduction system. It is a vital tool for measuring the heart rate variability (HRV), detecting any damage to the heart's muscle cells or any cardiac abnormality like myocardial ischemia and infarction, pre-excitation, atrial abnormalities, ventricular hypertrophy, pericarditis etc. [1–3]. It is also used to study the impact of various drugs on the heart. The required data, for diagnosing these ailments, is extracted from the four main parameters in an ECG signal, namely P-wave, QRS complex, T-wave and U-wave. Any change in the heart's structure or any abnormality in its functionality changes either the magnitude of these parameters or the distance between them.

The electrical potential of the heart's cell membranes are measured commonly using ten electrodes, placed on the chest and limbs of the patient. The acquired ECG signals get corrupted with various

kinds of noises, such as baseline wander (BW), power-line interference (PLI), motion artefacts, physiological artefacts etc. Among these, BW and PLI are the most prominent ones which deteriorate the signal quality and make the visual and automated diagnosis unreliable. BW is a low frequency noise, 0.05–0.7 Hz, arising mainly due to breathing or movement of the patient or due to any disturbance of electrodes [4,5]. On the other hand, PLI is a 50 or 60 Hz noise, added to ECG measuring devices from the power-line. Even if the device has very high common mode rejection ratio, this noise cannot be avoided completely. BW may disrupt the duration of ST segment, which is a crucial parameter in diagnosing various cardiovascular diseases [6]. Moreover, if the BW is large, it may lead to clipping of R peaks resulting in immeasurable HRV. PLI can mask the P and T waves of the ECG signal [7], rendering the signal useless or unreliable. Thus, BW and PLI must be suppressed from the recorded signal without removing any significant ECG attribute.

Over the years, various approaches have been explored to remove BW and PLI from ECG signals. Among these, the standard method to remove BW is by using a high pass filter [8] and a notch filter for PLI [9–11]. If implemented using finite impulse response (FIR) filters [8], computational delays and ringing effects are caused due to long responses, whereas infinite impulse response (IIR)

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filters can cause non-linear phase distortion [12,13]. These two issues have been addressed using zero phase bi-directional filters [14]; however the problem of tuning the bandwidth of the filter remains an issue. Also, a high pass filter used to remove BW may also discard some low frequency signal features; similarly a notch filter may remove the ECG signal component along with PLI.

In order to reject the time-varying noises and retain the ECG signal features of the same frequency range, adaptive filters have been proposed [15–18]. These adaptive filters require models of BW and PLI to generate reference signals. Authors in [19] modelled BW as first order fractional Brownian motion (fBm) process. While in [20], BW and PLI are jointly modelled using a combination of harmonically related sinusoids, each modulated by a low order polynomial. BW has also been estimated using cubic spline method [21], which assumes that the PR segment is well defined. The need of reference signals increases the computational complexity of the algorithm. Also, the adaptive algorithms including least mean squares (LMS) [22], normalized LMS [23] and recursive least squares (RLS) [15] suffer from low convergence speed and fixed step size. Modified versions of these algorithms [24–29] have been proposed to increase the convergence speed and in order to improve their performance in presence of noise.

Another approach is to consider signal decomposition techniques like empirical mode decomposition (EMD) and its variant algorithms [30–33], eigenvalue decomposition (EVD) [34], synchrosqueezed wavelet transforms (SSWT) [35] and variational mode decomposition (VMD) [36]. These techniques are very useful in dealing with non-stationary and non-linear real life signals. EMD algorithms [30,37–39] and their modified versions [40,41], discrete wavelet transforms [42–44] and EVD [5] have been used for ECG denoising. BW is present in several intrinsic mode functions (IMFs) or different levels of detail coefficients, thus removing BW in EMD-based methods leads to loss of significant features from ECG signal as well. In [5], authors proposed a denoising technique using eigenvalue decomposition of a Hankel matrix. It was noticed that the first and last eigenvectors correspond to the BW and PLI noises. These eigenvectors can, therefore, be removed in one step, removing the BW and PLI simultaneously. EVD methodology gives better results in comparison to existing methods available in the literature, however it is computationally complex. There are some studies in the literature for the removal of noise using Fourier coefficient thresholding techniques [45,46]. Moreover, the Fourier decomposition method (FDM) [47,48] has shown promising performance in problems such as nonlinear and nonstationary time-series decomposition, analysis and representation of signals in time-frequency-energy (TFE) domain [49–53]. In this paper, we propose a novel and efficient methodology to remove BW and PLI using FDM and compare the results with existing state-of-the-art techniques. We show that the proposed method clearly outperforms the other methods as the FDM decomposes a signal into desirable frequency bands with required cutoff frequencies, which facilitates effective BW and PLI removal.

Rest of the paper is organized as follows: Section 2 discusses some related works and the proposed methodology. Simulation results and discussions are presented in Section 3. Finally, conclusions are drawn in Section 4.

2. Methodology

2.1. Related work

In this section, we discuss some popular related works in this area as follows.

2.1.1. Empirical mode decomposition (EMD)

The EMD is a very popular method for analysis of non-stationary signals [30,32,38,39]. It decomposes a signal $x[n]$ into a set of intrinsic mode functions (IMFs) $y_i[n]$ and a residue $r[n]$, i.e.,

$$x[n] = \sum_{i=1}^L y_i[n] + r[n] \quad (1)$$

with L denoting the number of IMFs. These IMFs must satisfy the following conditions [30,32,38,39]:

- The number of extremas (minima and maxima) and the number of zero crossings in the entire duration of the signal, are equal or at most differ by one.
- The average of upper and lower envelopes obtained by interpolation of local maxima and minima, is zero for any time in the complete duration of the signal.

After decomposing the signal into IMFs, relevant features are extracted and used for classification (or denoising) of ECG signals [37,40,41].

2.1.2. Eigenvalue decomposition (EVD)

Let $x[n]$ denote a real multi-component non-stationary signal. A Hankel matrix $H^x = [H]_{m_1, m_2}$, for $m_1, m_2 = 1, 2, \dots, L$, is formed using sample values from $x[n]$, such that H_{m_1, m_2} is constant for a given value of $(m_1 + m_2)$. The eigenvalue decomposition of this matrix is performed as [5]

$$H^x = V_x \Lambda_x V_x', \quad (2)$$

where V_x is a real matrix comprising eigenvectors corresponding to the L eigenvalues stored as diagonal entries in Λ_x . When the signal is corrupted with BW and PLI, the end-point eigenvalues are most affected while there is a negligible change in all the other eigenvalues. The corrupted eigenvalues are thus identified and removed before reconstructing the signal using healthy eigenvalues. In this manner, the noise component is reduced in the signal.

2.1.3. Normalized sign least mean square (NSLMS)

In this method, least mean square adaptive filter is used. The weights of the filter are updated as [54]

$$\mathbf{w}[n+1] = \mathbf{w}[n] + \mu[n]\mathbf{x}[n]\text{sgn}\{e[n]\}, \quad (3)$$

where $\text{sgn}\{\}$ denotes the signum function, $\mu[n]$ and $e[n]$ represent the variable step size and the error, respectively. Noisy signal is applied as the desired response of the LMS filter and a noise signal $p[n]$ recorded from another noise generator that is in some way correlated with the noise corrupting the signal, is applied as input to the filter. The mean square error (MSE) is thus obtained as

$$E\{e^2[n]\} = E\{(x[n] - y[n])^2\} + E\{p^2[n]\}, \quad (4)$$

where $E\{\}$ is the expectation operator and filter output $y[n] = \mathbf{w}^T[n]\mathbf{x}[n]$. Minimizing the MSE yields an output which is a least square estimate of the original signal.

2.2. Proposed method

The FDM is an adaptive signal decomposition technique which divides a given signal (or data) into a set of Fourier intrinsic band functions (FIBFs) that are local, adaptive, complete and energy-preserving. The FDM preserves the energy of a signal $x[n]$ by decomposing it into a set of orthogonal or linearly independent non-orthogonal yet energy preserving (LINOEP) FIBFs with desir-

able frequency bands using the following signal decomposition model [47,48,55,51]:

$$x[n] = c_0 + \sum_{l=1}^M x_l[n] = c_0 + \sum_{l=1}^M c_l[n], \quad (5)$$

where c_0 is the average value of signal $x[n]$, $x_l[n]$ and $c_l[n]$ denote M orthogonal and LINOEP FIBFs, respectively, with $l = 1, 2, \dots, M$.

A block diagram illustrating FDM technique using DFT and DCT based zero-phase filter-bank (i.e. $H_l[k] \in \mathbb{R}_{\geq 0}$, $\forall l, k$) is shown in Fig. 1, where the signal $x[n]$ is decomposed into a set of desired orthogonal components $\{x_1[n], x_2[n], \dots, x_M[n]\}$. For the DFT based FDM [55], the frequency response of l th band in the filter-bank can be obtained by defining $H_l[k] = 1$ for the desired band of frequencies and zero otherwise, and these filters can be expressed as follows:

$$H_l[k] = \begin{cases} 1, & (K_{l-1} + 1) \leq k \leq K_l \\ & \& (N - K_l) \leq k \leq (N - K_{l-1} - 1), \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where $l = 1, 2, \dots, M$, $K_0 = 0$ and $K_M = N/2$ (consider $(N - 1)/2$ in case $N/2$ is not integer). Using inverse DFT (IDFT) operation, the signal components $x_l[n]$ are obtained as

$$x_l[n] = \sum_{k=0}^{N-1} [H_l[k] X[k] \exp(j2\pi kn/N)], \quad (7)$$

where N length DFT of signal $x[n]$ is defined as follows:

$$X[k] = \frac{1}{N} \sum_{n=0}^{N-1} x[n] \exp(-j2\pi kn/N), \quad 0 \leq k \leq N - 1. \quad (8)$$

For the DCT based FDM [51], filters $H_l[k]$ are expressed as

$$H_l[k] = \begin{cases} 1, & (K_{l-1} + 1) \leq k \leq K_l, \\ 0, & \text{otherwise} \end{cases} \quad (9)$$

with $K_0 = 0$ and $K_M = N$. Using the inverse discrete cosine transform (IDCT) type-2, $x_l[n]$ are obtained as follows:

$$x_l[n] = \sqrt{\frac{2}{N}} \sum_{k=0}^{N-1} \sigma_k H_l[k] X_{c2}[k] \cos\left(\frac{\pi k(2n+1)}{2N}\right), \quad (10)$$

where DCT for the signal $x[n]$ is defined as

$$X_{c2}[k] = \sqrt{\frac{2}{N}} \sum_{n=0}^{N-1} x[n] \cos\left(\frac{\pi k(2n+1)}{2N}\right), \quad 0 \leq k \leq N - 1, \quad (11)$$

with N denoting the length of the signal, and normalization factors $\sigma_k = \frac{1}{\sqrt{2}}$ for $k = 0$, and $\sigma_k = 1$ for $k \neq 0$.

In this study, we use FDM to decompose the ECG signal into three FIBFs, where first component ($x_1[n]$) is BW noise, second one ($x_2[n]$) is PLI, and last one ($x_3[n]$) is the clean ECG signal. To obtain the three components, one DFT (DCT) and three IDFT (IDCT) operations are required, where all these operations are implemented using the fast Fourier transform (FFT) algorithm. Practically, the first two components would be discarded and only noise-free ECG signal is passed for further processing and identification of various cardiac anomalies. Motivated by this requirement, we propose the following single stage filters:

$$H[k] = \begin{cases} 0, & 0 \leq k \leq K_1 \quad \& (N - K_1) \leq k \leq (N - 1), \\ 0, & K_2 \leq k \leq K_3 \quad \& (N - K_3) \leq k \leq (N - K_2), \\ 1, & \text{otherwise} \end{cases} \quad (12)$$

for DFT based filtering and

$$H[k] = \begin{cases} 0, & 0 \leq k \leq K_1, \\ 0, & K_2 \leq k \leq K_3, \\ 1, & \text{otherwise} \end{cases} \quad (13)$$

for DCT based filtering, where K_1 corresponds to BW cutoff frequency, K_2 and K_3 are lower and upper cutoff frequencies corresponding to PLI signal. This is an efficient implementation because it requires only one DFT (DCT) and one inverse DFT (DCT) to remove BW and PLI simultaneously. Since the computation of DFT (DCT) is based on fast Fourier transform (FFT) algorithm, the computational complexity of the proposed algorithm is $\mathcal{O}(N \log_2 N)$.

The zero-phase filtering in the proposed method is achieved by using non-negative real values for defining the response of each filter. This helps in preserving the salient features of the ECG signal, including the maxima and minima, i.e., P, Q, R, S, and T waves, precisely at the same position where these complexes or features exist in the original waveform. On the contrary, the conventional non-zero-phase filtering techniques result in shifting these complexes in the filtered signal and are thus not suitable for obtaining meaningful information from the given ECG data.

3. Simulation results and discussions

The performance of the proposed method has been evaluated on the MIT-BIH Arrhythmia database [56–58]. 16 ECG signals, each with a recording of 30 min, have been considered in this work. A sample of real BW noise, recorded over 30 min, has also been taken from the same database. PLI has been generated as a sinusoidal signal of unit amplitude and 60 (or 50) Hz frequency. Noisy signal is obtained by adding BW and/or PLI to the clean signal at four different levels of input SNR (SNR_i), i.e., 5 dB, 0 dB, –5 dB, and –10 dB, respectively. For the purpose of easier computation, the signal is divided into blocks of 30 s. Each block is filtered using the proposed method, and then these blocks are recombined to obtain the complete signal of 30 min. Fig. 2 demonstrates the step-by-step procedure for the proposed methodology.

The efficiency of our method is compared to the existing state-of-the-art techniques, in terms of percent root mean square difference (PRD), computed as follows [59]:

$$\text{PRD} = 100 \sqrt{\frac{\sum_{n=1}^N (x[n] - \hat{x}[n])^2}{\sum_{n=1}^N x^2[n]}}, \quad (14)$$

where $x[n]$ and $\hat{x}[n]$ represent the original noise-free ECG signal and the estimated signal obtained after filtering the noisy signal, respectively. For a 30 min recording, sampled at $f_s = 360$ Hz, we have a total of 6,48,000 samples, i.e. $N = 6,48,000$. With $(x[n] - \hat{x}[n])$ indicating the noise after filtering, the output SNR is obtained as

$$\text{SNR}_o = 10 \log_{10} \left(\frac{\sum_{n=1}^N \hat{x}^2[n]}{\sum_{n=1}^N (x[n] - \hat{x}[n])^2} \right). \quad (15)$$

The strength of the proposed method is highlighted in Fig. 3, where the filtered signal (Fig. 3(e)) resembles the clean signal (Fig. 3(a)) quite closely, resulting in a very small difference signal (Fig. 3(h)), even at a poor input SNR, i.e., a combined noise containing BW and PLI, each having a power 10 dB more than the clean signal. Moreover, the proposed method clearly distinguishes the two noises, BW (Fig. 3(b)) and PLI (Fig. 3(c)) present in the noisy signal (Fig. 3(d)), and separates them effectively (Fig. 3(f) and (g)). As shown in the illustration, large levels of noise, with a power 10 times the clean signal, are also easily isolated using the proposed method. The results only get better as we increase the input SNR.

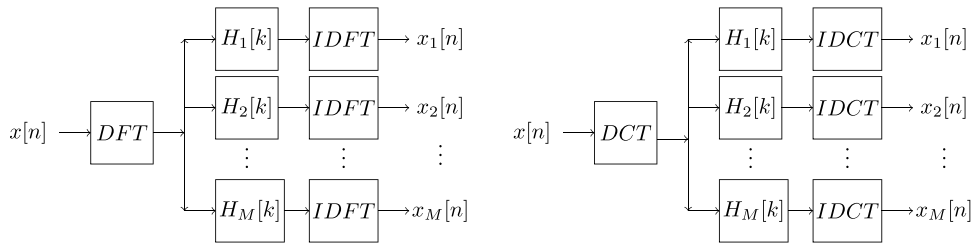


Fig. 1. Block diagram for FDM technique based on DFT and DCT to decompose a signal into orthogonal components with desirable frequency bands.

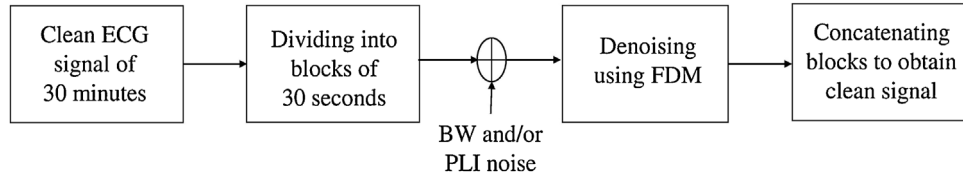


Fig. 2. Step-by-step procedure for obtaining a clean signal from a noisy ECG signal using FDM.

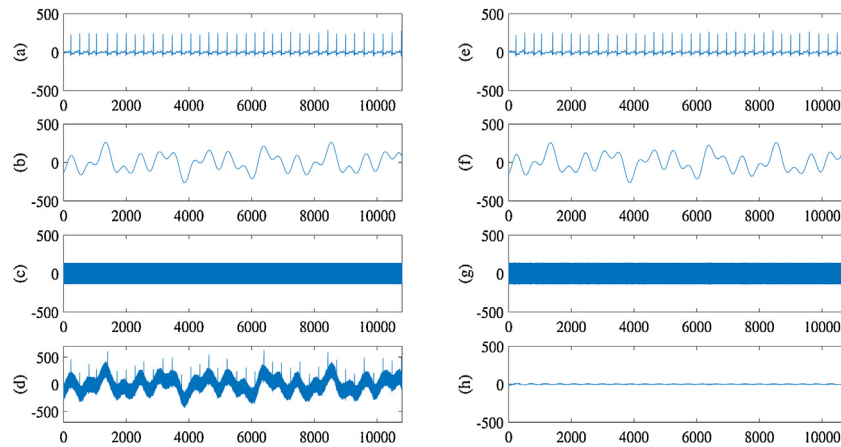


Fig. 3. Plots of ECG signal: (a) clean ECG signal, (b) 10 dB BW noise, (c) 10 dB PLI at 60 Hz frequency, (d) noisy signal corrupted with both BW and PLI, each of the two noises having 10 dB more power than the original signal, (e) filtered ECG signal obtained using FDM, (f) BW extracted from noisy signal, (g) PLI extracted from noisy signal, (h) difference signal obtained by subtracting filtered signal from the clean signal.

3.1. Power-line interference

Considering power-line interference of 60 Hz, Fig. 4 compares the PRD values obtained by the proposed method with the existing techniques like eigenvalue decomposition (EVD) [5], empirical mode decomposition-wavelet transform (EMD-WT) method discussed in [40], modified recursive least square (MRLS) based method [60] and normalized sign least mean square (NSLMS) [54]. The performance of NSLMS method degrades the most as the input SNR reduces. Although MRLS method performs slightly better than the NSLMS method, it is also vulnerable to increasing levels of noise. EMD-WT method is shown to exhibit some resistance to change in input SNR. The recently proposed EVD method produces good results at low levels of noise, but fails to maintain the trend with increase in noise power. In comparison to all these methods, our method is the least affected by change in noise power and obtains the lowest PRD values (best in class results) at all four levels of input SNR. As shown in Fig. 4, the performance of the proposed method at an input SNR of -10 dB is remarkably better than the existing techniques. Similar performance is also observed for PLI of 50 Hz.

Table 1 compares different methods in terms of output SNR values (SNR_o) achieved for various input SNR (SNR_i). While EMD-WT is able to produce decent SNR_o when $SNR_i = -10$ dB, it fails to

offer significant improvement at higher input SNR, i.e., $SNR_i = 5$ dB. On the other hand, EVD method outperforms EMD-WT at $SNR_i = 5$ dB, but it suffers from sharp variations as we change SNR_i . In contrast to the existing methods, the proposed method produces better SNR_o at all levels of SNR_i . For example, we achieve $SNR_o = 27.6$ dB for record 100 at $SNR_i = -10$ dB as opposed to $SNR_o = 19$ dB and $SNR_o = 12.5$ dB obtained by EVD and EMD-WT methods, respectively.

3.2. Baseline wander noise

Next, we consider the role of BW noise. It occurs at very low frequencies, typically less than 0.7 Hz [5]. The performance of the proposed method in removing BW noise from ECG signals is compared with the existing methods including EVD [5], another method [61] based on extended Kalman filter (EKF), NSLMS method [54], and Hilbert vibration decomposition (HVD) based method [62]. Fig. 5 shows the PRD values obtained by these methods for the 16 ECG signals considered in this work, corrupted by BW noise at input SNR of 5 dB, 0 dB, -5 dB, and -10 dB, respectively. HVD based method performs inferior to the other methods and is strongly affected by drop in the SNR values. The PRD values obtained by NSLMS method increase gradually with increase in power of noise. The noise removal techniques based on EKF and EVD exhibit decent

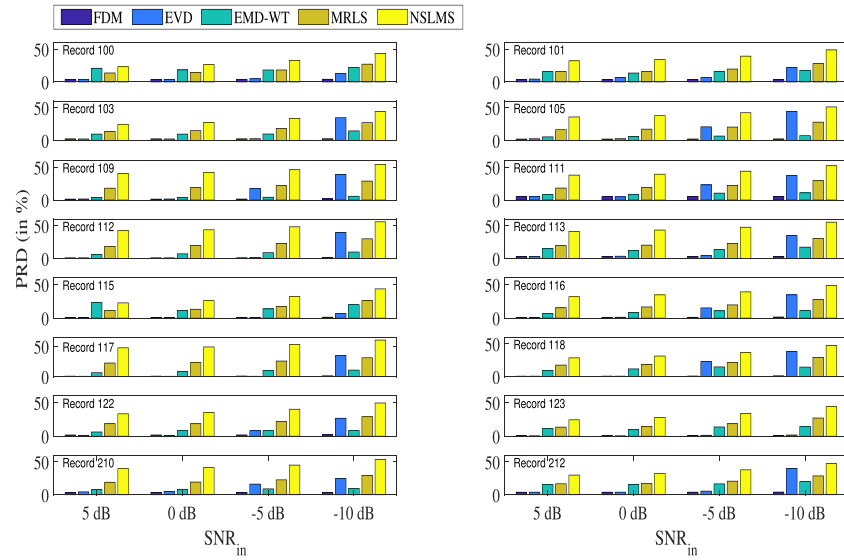


Fig. 4. PRD values obtained by the proposed method as compared to the existing methods, considering 60 Hz PLI.

Table 1

Output SNR values obtained after removing 60 Hz PLI from ECG signal at different values of input SNR.

ECG record	SNR_i	Proposed method				EVD method				EMD-WT method			
		-10 dB	-5 dB	0 dB	5 dB	-10 dB	-5 dB	0 dB	5 dB	-10 dB	-5 dB	0 dB	5 dB
100	SNR_o	27.6	27.8	27.9	28.1	19.0	27.5	28.1	28.1	12.5	14.4	14.5	13.3
101		28.6	29.0	29.1	29.1	14.2	27.7	27.7	29.1	14.7	15.9	17.2	16.2
103		31.0	31.5	31.7	31.7	8.3	32.3	33.0	33.0	15.6	19.5	20.1	19.8
105		31.9	32.6	32.9	33.0	6.4	14.0	31.9	32.6	22.4	23.8	24.4	25.4
109		35.0	36.2	36.7	36.9	6.9	17.9	35.3	35.3	24.1	26.3	27.3	28.1
111		25.4	25.6	25.7	25.7	5.0	15.7	25.2	25.7	19.0	19.6	21.1	21.2
112		35.0	36.9	37.7	38.0	6.8	40.2	41.8	41.8	19.7	20.7	23.2	23.6
113		29.7	30.0	30.1	30.2	8.4	28.2	29.9	29.9	14.7	16.9	17.8	16.3
115		36.9	39.9	41.5	42.2	31.3	45.9	46.0	46.0	13.6	16.9	18.6	12.6
116		35.5	37.8	38.9	39.3	9.7	23.5	43.0	43.0	18.6	19.2	21.6	22.8
117		36.4	39.0	40.3	40.8	8.5	42.3	42.4	42.4	19.2	19.6	21.3	23.5
118		37.1	39.7	41.0	41.6	14.3	15.9	45.4	45.5	16.6	16.8	18.6	20.6
122		33.4	34.3	34.6	34.7	14.9	46.4	35.3	35.3	20.8	21.6	21.9	23.9
123		37.9	41.0	42.8	43.6	42.7	45.0	47.2	47.2	16.5	16.9	19.8	19.2
210		28.5	28.7	28.8	28.9	16.3	17.2	28.1	28.1	20.6	21.5	21.8	22.3
212		27.7	28.0	28.0	28.1	6.8	26.0	28.3	28.3	13.8	15.5	16.2	16.0

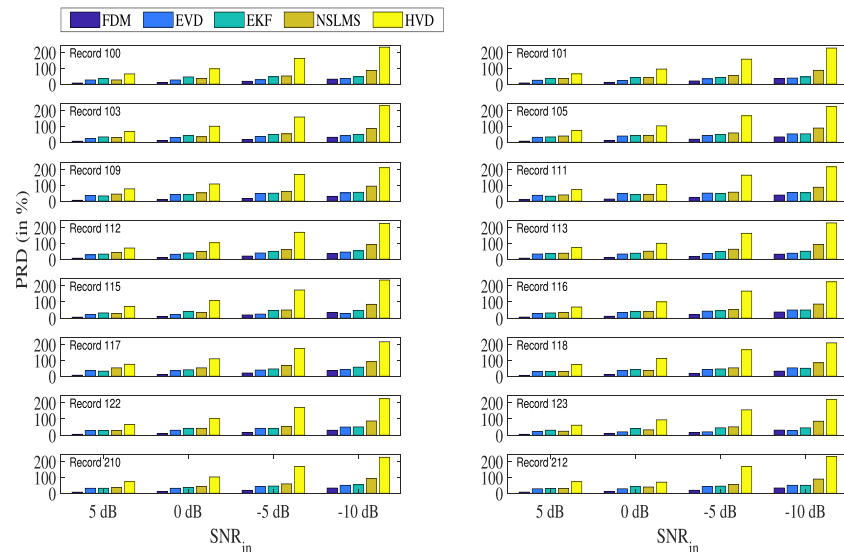
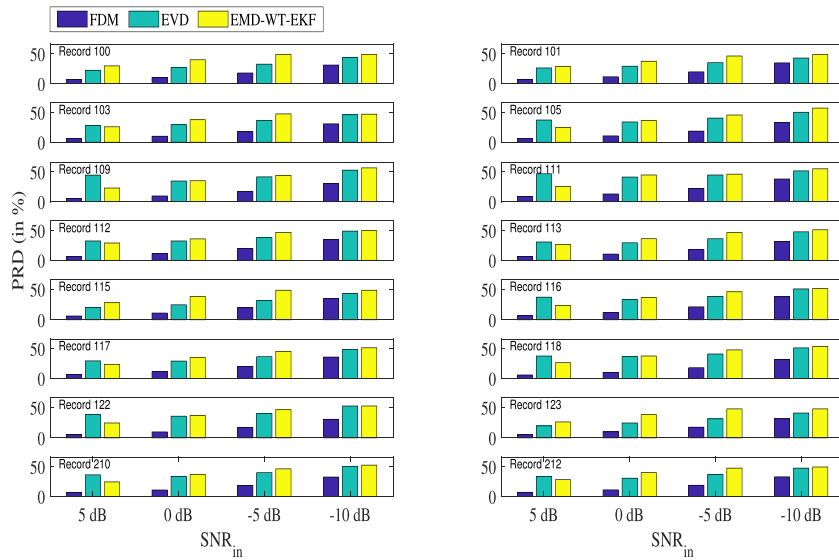


Fig. 5. PRD values obtained by the proposed method as compared to the existing methods, considering BW noise.

Table 2

Output SNR values obtained after removing BW noise from ECG signal at different values of input SNR.

ECG record	SNR _i	Proposed method				EVD method				EKF method			
		−10 dB	−5 dB	0 dB	5 dB	−10 dB	−5 dB	0 dB	5 dB	−10 dB	−5 dB	0 dB	5 dB
100	SNR _o	10.5	15.1	19.4	23.0	8.4	10.9	12.6	13.1	3.9	7.1	10.2	11.3
101		19.7	14.2	18.9	23.1	7.6	9.8	11.4	11.8	4.0	6.8	9.2	10.6
103		10.6	15.4	20.2	24.9	6.7	8.7	10.9	11.9	6.0	7.1	8.2	10.2
105		10.1	14.8	19.6	24.3	3.5	5.7	7.4	9.1	4.1	6.3	8.8	9.9
109		10.7	15.5	20.4	25.3	3.4	4.8	6.5	8.4	4.1	6.6	8.3	9.3
111		9.1	13.7	18.5	23.3	3.2	4.1	5.3	7.5	4.3	6.9	8.6	9.9
112		9.7	14.3	19.1	23.9	4.9	7.2	9.3	10.3	3.4	5.6	7.2	8.0
113		10.5	15.1	19.8	24.5	7.1	8.6	9.6	9.6	3.7	6.0	7.7	8.5
115		9.7	14.3	19.1	24.0	10.1	11.8	13.6	13.3	7.6	7.3	8.2	10.2
116		9.1	13.7	18.5	23.3	4.2	6.4	8.7	10.2	4.0	6.8	9.2	10.6
117		9.4	14.1	19.0	23.9	6.4	7.9	8.9	8.9	3.4	5.4	6.7	7.3
118		10.3	15.0	19.9	24.6	3.1	5.2	7.7	9.3	3.4	6.6	9.2	10.8
122		10.7	10.4	20.3	25.1	3.5	5.7	8.3	10.0	3.9	6.7	9.0	10.3
123		10.5	15.3	20.1	25.0	10.7	12.8	13.8	13.2	4.1	7.2	10.3	11.3
210		10.2	14.9	19.7	24.4	4.2	6.2	8.7	9.8	4.0	6.5	8.3	9.3
212		10.1	14.7	19.3	23.5	4.7	7.0	9.6	10.8	4.0	6.9	9.5	10.6

**Fig. 6.** PRD values obtained by the proposed method as compared to the existing methods, considering both BW and PLI.

results but fail to maintain a uniform trend across different ECG signals. Although an increase in the power of noise adversely affects the performance of these methods, the proposed method produces small PRD values even at lowest input SNR of −10 dB. For all the ECG signals, our method clearly outperforms the existing techniques at different levels of SNR.

In terms of output SNR also, the proposed method obtains better results in comparison to its counterparts, as shown in Table 2. SNR_o increases gradually with increase in SNR_i for all the methods. As the noise power decreases than the signal power, i.e., SNR_i ≥ 0 dB, our method performs significantly better than EVD and EKF methods, producing output SNR in the range 18.5 dB < SNR_o < 20.4 dB for SNR_i = 0 dB and 23.0 dB < SNR_o < 25.3 dB for SNR_i = 5 dB.

3.3. Both baseline wander noise and power-line interference

We finally consider the combined effect of both the BW noise as well as PLI. It is difficult to simultaneously remove both the anomalies from the noisy signal, hence, there do not exist many methods for comparison. EVD method [5] is amongst the first to produce such results. For the sake of comparison, we use EMD-WT method [40] for canceling power-line interference, while EKF method [61] is used to remove BW noise. Bar plots of PRD values

obtained by our method are compared with EVD and EMD-ET-EKF methods in Fig. 6 for 16 ECG signals at different levels of SNR_i. While EVD method produces good results for SNR_i ∈ {0, −5, −10} dB, it performs inferior to the EMD-WT-EKF method at 5 dB SNR_i. The PRD values for all the methods increase as the input SNR reduces, however, the proposed method produces the lowest PRD values at different levels of SNR_i. The results strongly validate the superior performance achieved by our method in removing both the noises together for all the ECG signals considered here.

Table 3 lists the SNR_o values for our method in comparison to the existing methods at different levels of SNR_i. For record 100, SNR_o obtained by our method at SNR_i = 5 dB is better than EVD method and EMD-WT-EKF method by 10.4 dB and 13 dB, respectively. Moreover, the proposed method shows a larger promise to be used in real-world scenarios as it produces consistent results across all the ECG records considered in this work.

4. Conclusion and future scope

In this work, we have proposed a robust and efficient method to de-noise a recorded ECG signal by suppressing both BW and PLI.

Table 3

Output SNR values obtained after removing both BW and PLI from ECG signal at different values of input SNR.

ECG record	SNR _i	Proposed method				EVD method				EMD-WT-EKF method			
		−10 dB	−5 dB	0 dB	5 dB	−10 dB	−5 dB	0 dB	5 dB	−10 dB	−5 dB	0 dB	5 dB
100	SNR ₀	10.5	15.1	19.5	23.3	7.9	9.9	11.5	12.9	6.0	6.6	8.3	10.3
101		9.6	14.2	18.8	23.0	8.1	9.4	10.8	11.8	5.6	6.8	8.8	11.3
103		10.6	15.3	20.0	24.4	7.5	9.1	10.9	11.0	6.3	7.1	9.1	12.2
105		10.1	14.7	19.5	24.1	6.5	8.0	9.4	6.0	5.7	7.1	9.3	12.8
109		10.7	15.4	20.3	25.1	6.1	7.8	9.9	5.3	5.3	7.1	9.4	12.9
111		9.0	13.4	17.8	21.4	6.7	7.6	8.2	5.9	5.2	6.9	7.6	12.4
112		9.7	14.3	19.2	24.1	7.0	8.7	10.0	9.4	5.4	6.9	9.3	12.8
113		10.5	15.1	19.7	24.0	7.2	9.0	10.9	10.1	4.9	6.7	9.0	11.6
115		9.6	15.3	19.2	24.1	8.3	10.4	12.1	13.7	6.3	6.9	8.9	13.3
116		9.1	13.7	18.5	23.4	6.7	8.5	9.6	8.2	6.0	7.0	9.2	12.6
117		9.4	14.1	19.0	23.9	7.0	9.1	10.6	10.4	4.7	6.6	9.1	12.7
118		10.4	15.1	20.0	24.9	6.6	8.5	9.0	5.9	5.9	6.8	8.9	12.4
122		10.7	15.4	20.2	24.9	6.4	8.2	9.8	6.6	5.9	7.1	9.2	12.3
123		10.5	15.2	20.2	25.1	8.6	10.5	12.4	14.0	6.3	7.0	9.0	12.2
210	SNR ₀	10.2	14.8	19.3	23.4	6.6	8.2	9.5	7.1	5.4	7.0	9.2	12.5
212		10.1	14.7	19.2	23.1	7.3	8.8	10.4	9.2	5.6	6.6	8.6	11.1

The well-known DFT and DCT based Fourier decomposition method (FDM) is used to decompose the noisy signal into a set of orthogonal Fourier intrinsic band functions. Thereafter, the frequency components arising out of the noise sources are identified carefully to avoid any overlap with useful information in the signal. We have designed optimal filters to remove these components—BW and PLI, while preserving the actual peaks of the clean ECG signal. The simulation results obtained on the MIT-BIH Arrhythmia database clearly highlight the strength of our method as compared to the existing state-of-the-art techniques over a range of SNR values. The proposed method is a strong contender to be used as a pre-processing step before analysing ECG signals for diagnosis of various diseases. Furthermore, it can be implemented in real-time applications owing to its high efficiency and accuracy.

Our future direction of research would include the following: We would like to compute complexity of the existing algorithms to provide a comparison of different techniques. We would also include some studies catering to the results inferred from the denoised signals, in terms of noting the R peaks to measure heart rate variability to detect different related diseases. We would further adapt the proposed method to estimate and remove other sources of noise in the ECG signals and perform the mean opinion score (MOS) test by collaborating with medical experts in this field. Moreover, the denoised signal obtained by our method would be used for extracting relevant features to detect and classify different heart abnormalities.

Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

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